



Department of Mechanical Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE 'B' Section

Course Code & Title: ME8792 Power Plant Engineering

Name of the Faculty member (s): Mr.P.Pavithran , AP/Mech

Innovative Practice Description

- **Unit / Topic:** Unit IV / Virtual Simulation of Renewable Energy Power Plant
- **Course Outcome:** Explain the layout, construction and working of the components inside Renewable energy power plants.
- **Topic Learning Outcome:**
 - ✓ Classify hydroelectric power plant
 - ✓ Explain the working of the different renewable energy systems
 - ✓ Draw the layout of different renewable energy systems
 - ✓ Explain the working of different types of fuel cell and its applications
- **Activity Chosen:** Virtual laboratory
- **Justification:**
 - ✓ Apply basic engineering concept in Renewable energy systems
 - ✓ Explain the advantages of renewable energy system based on environmental issue
 - ✓ Apply the fundamental of renewable energy sources in project
 - ✓ Explain the working of different renewable energy systems.
- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

Totally 30 students were actively participated for this online virtual simulation practice. Our nation has abundant solar, water, wind, and geothermal energy resources, and many U.S. companies are developing, manufacturing, and installing cutting-edge, high-tech renewable energy systems. The Office of Energy Efficiency and Renewable Energy (EERE) leads a large network of researchers and other partners to deliver innovative technologies that will make renewable electricity generation cost competitive with traditional sources of energy. Working with our national laboratories and through these partnerships, we are catalyzing the transformation of the nation's energy system and building on a tradition of U.S. leadership in science and engineering as a cornerstone of our economic prosperity.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
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- **PO / PSO mapped:**

CO	PSO1	PSO2	PSO3
C206.4	1		

- **Images / Screenshot of the practice:**

You are the manager of the Virtual Power Plant and in charge of preventing a blackout.

The frequency of the power grid, displayed at the top of the screen, needs to stay between 49.9 hertz and 50.2 hertz to prevent the power grid from collapsing. Grid frequency is in a constant state of fluctuation, which is increased by unpredictable incidents. Use the four controllers to adjust power production at the decentralized power plants and rebalance grid frequency. If the frequency decreases, power production needs to rise or demand needs to fall. If frequency rises, try reducing power production or increasing the demand for power. **Quick tip:** At the bottom edge of the screen, you can switch from manual mode to VPP mode to see how the automatic control system in a Virtual Power Plant works.



Figure1. “Illustrate the operation of renewable energy power plant”

- **Reflective Critique:**

❖ *Feedback of practice from students and other stakeholders:*

The overall feedback from the students were really good and students felt that it is very easy understand the real time operation of renewable energy power plant

❖ ***Benefit of the practice:***

- ✓ After completing this hand on session in virtual laboratory, the students were able to understand the operation of renewable power plant very clearly.
- ✓ This virtual simulation enables the students to better understanding of their real time operation of renewable energy power plants such as Solar and wind by adjusting the frequency of the system to grid.
- ✓ This virtual simulation also helps the students to know the stability of the system.

❖ ***Challenges faced in implementation:***

The output of simulation part can be better.

References:

- ❖ <https://www.next-kraftwerke.com/virtual-power-plant-vpp-simulation/?lang=en>

Signature of Faculty Member

HOD